

TECHNICAL EVALUATION EXAMINATION

Core Algorithms & Data Structures (3-Question Problem Set)

Course / Topic: Advanced Algorithmic Design

Total Marks: 100 Marks

Duration: 180 Minutes (3 Hours)

Date: July 17, 2026

INSTRUCTIONS TO CANDIDATES:

- Answer all questions. Ensure code solutions are optimal in both time and space complexity.
- State any assumptions made clearly alongside your analysis.
- Include runtime complexity analysis (Big-O notation) for all implementations.

Question 1: Recursion & Formal Verification

[30 Marks]

Problem: The McCarthy 91 Function

The McCarthy 91 function is a classic nested-recursive function designed by computer scientist John McCarthy. It is widely used in formal verification as a benchmark for proving termination of complex recursive schemes.

Write a recursive function $M(n)$ that takes an integer n as input and computes the output based on the following mathematical definition:

$$M(n) = \begin{cases} n - 10 & \text{if } n > 100 \\ M(M(n + 11)) & \text{if } n \leq 100 \end{cases}$$

EXAMPLES

Input: $n = 99$
Output: 91
Explanation: $M(99) = M(M(110)) \rightarrow M(100) \rightarrow M(M(111)) \rightarrow M(101) \rightarrow 91$

Input: $n = 105$
Output: 95
Explanation: Since $105 > 100$, return $105 - 10 = 95$.

Constraints:

- $-2 \times 10^9 \leq n \leq 2 \times 10^9$

Question 2: Advanced Binary Search (Search on Answer Space)

[35 Marks]

Problem: Magnetic Force Between Two Balls (LeetCode 1552)

In the universe of Rick and Morty, Morty has n empty baskets placed at discrete 1D physical coordinate points. The coordinate position of the i -th basket is given by the array element $position[i]$.

Rick wants to distribute m identical magnetic balls into these baskets such that the **minimum magnetic force** between any two nested balls is **maximized**. The magnetic force between any two balls located at coordinate positions x and y is calculated as the absolute difference: $|x - y|$.

EXAMPLES

Input: `position = [1, 2, 3, 4, 7]`, `m = 3`

Output: 3

Explanation: Distributing 3 balls at positions 1, 4, and 7 gives a minimum magnetic force of 3 (since $|1-4|=3$, $|4-7|=3$, and $|1-7|=6$).

Input: `position = [5, 4, 3, 2, 1, 1000000000]`, `m = 2`

Output: 999999999

Explanation: Placing balls at positions 1 and 1000000000 maximizes the minimum distance.

Constraints:

- $2 \leq position.length \leq 10^5$
- $1 \leq position[i] \leq 10^9$
- All integers in $position$ are distinct.
- $2 \leq m \leq position.length$

Question 3: Sliding Window Technique

[35 Marks]

Problem: Longest Repeating Character Replacement (LeetCode 424)

You are given a string s consisting of uppercase English letters and an integer k . You are allowed to choose any character of the string and change it to any other uppercase English character. You can perform this substitution operation at most k times.

Determine and return the *maximum length* of a contiguous substring containing all identical characters that can be achieved after applying these operations.

EXAMPLES

Input: $s = \text{"ABAB"}, k = 2$

Output: 4

Explanation: Replace both 'A's with 'B's to yield "BBBB" (or vice versa to yield "AAAA").

Input: $s = \text{"AABABBA"}, k = 1$

Output: 4

Explanation: Replace the middle 'A' with 'B' to get "AABBBBA". The subsegment "BBBB" has the longest length of 4.

Constraints:

- $1 \leq s.length \leq 10^5$
- s consists of uppercase English letters only.
- $0 \leq k \leq s.length$